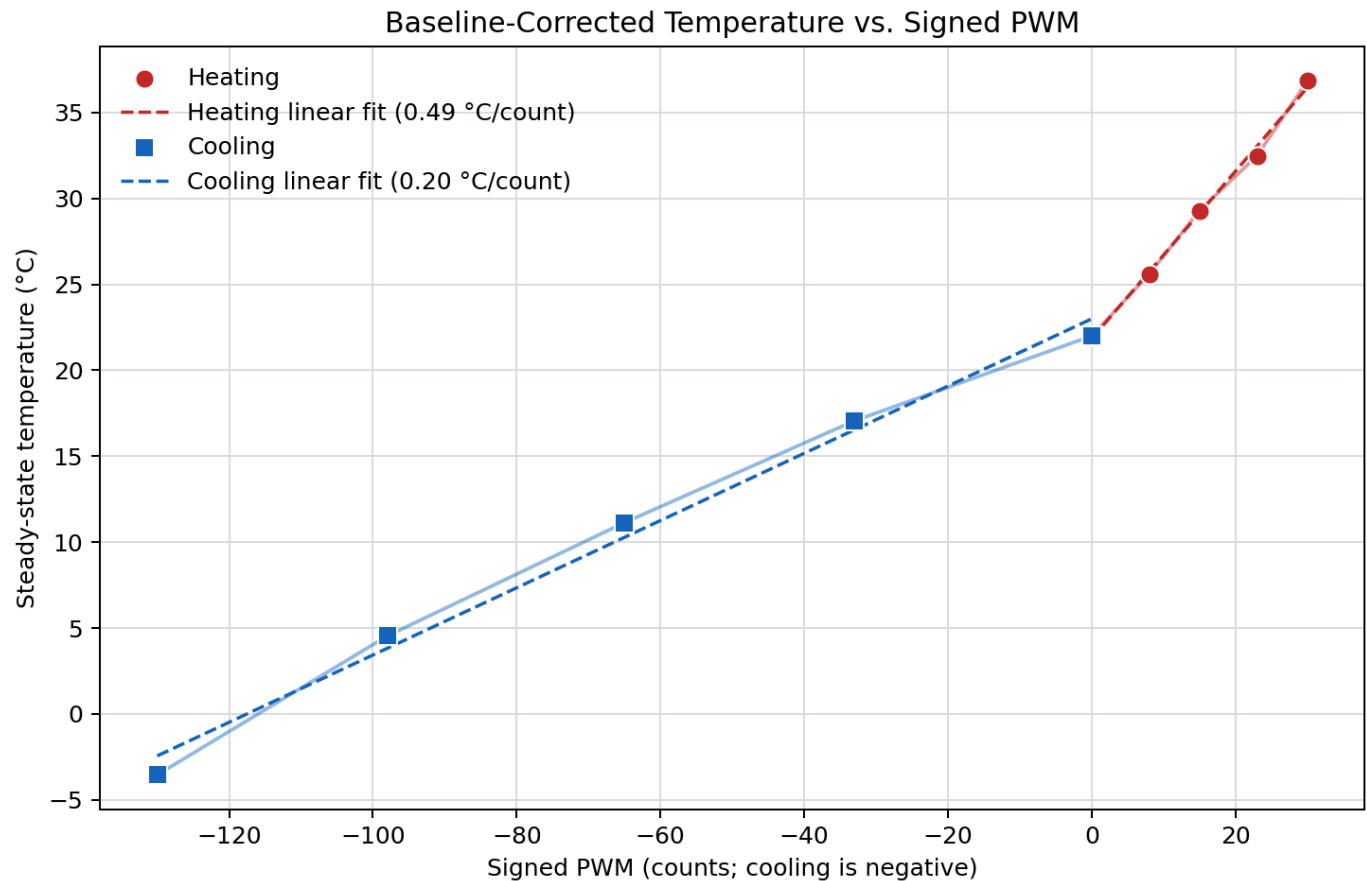


Kadener-Wijnaendts-Module4

A2: TEC Heating and Cooling Analysis

1. Part 4: Combined Graph and Fits



2. Part 5.1: Measured Slopes

The fitted heating and cooling-slope magnitudes are

$$m_h \approx 0.49 \frac{^{\circ}\text{C}}{\text{PWM count}}, \quad m_c \approx 0.20 \frac{^{\circ}\text{C}}{\text{PWM count}}.$$

Their measured ratio is

$$r = \frac{m_h}{m_c} = \frac{0.49}{0.20} \approx 2.45.$$

Thus, one additional heating PWM count changes the steady temperature about 2.45 times as much as one additional cooling PWM count in magnitude. The visible cooling curvature makes the fitted value somewhat dependent on the selected range, but the single fit captures the main heating-cooling asymmetry.

3. Part 5.2: PWM and the Slope-Ratio Model

For PWM duty cycle D , the current equals I during the on-time $D\tau$ and zero during the remainder of a period τ . Therefore,

$$\langle I \rangle = \frac{1}{\tau} \int_0^\tau I(t) dt = \frac{I(D\tau)}{\tau} = DI,$$

and

$$\langle I^2 \rangle = \frac{1}{\tau} \int_0^\tau I^2(t) dt = \frac{I^2(D\tau)}{\tau} = DI^2.$$

Hence, both the Peltier contribution and Joule contribution scale linearly with duty cycle for a fixed on-state current. Let $\dot{Q}_P$ and $\dot{Q}_J$ be their full-duty object-face magnitudes, and let G be the passive thermal conductance to the surroundings. At steady state,

$$G(T_h - T_0) = D(\dot{Q}_P + \dot{Q}_J)$$

for heating, while the cooling temperature-change magnitude obeys

$$G|T_c - T_0| = D(\dot{Q}_P - \dot{Q}_J).$$

Since $D = |u|/255$, the corresponding slope magnitudes are

$$m_h = \frac{\dot{Q}_P + \dot{Q}_J}{255G}, \quad m_c = \frac{\dot{Q}_P - \dot{Q}_J}{255G}.$$

Their ratio is therefore

$$r = \frac{m_h}{m_c} = \frac{\dot{Q}_P + \dot{Q}_J}{\dot{Q}_P - \dot{Q}_J}.$$

Writing $x = \dot{Q}_J/\dot{Q}_P$ gives $r = (1 + x)/(1 - x)$. Solving,

$$r - rx = 1 + x, \quad r - 1 = x(r + 1),$$

so

$$\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{r - 1}{r + 1}.$$

Using the measured ratio,

$$\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{2.45 - 1}{2.45 + 1} = \frac{1.45}{3.45} \approx 0.42.$$

Thus, the object-face Joule heating is approximately 42% of the Peltier heat-transfer magnitude in this model.

4. Part 5.3: Laird Datasheet Calculation

The values come from the Specifications table on page 3 of the datasheet, using the column for a hot-side temperature of 27°C .

- $R_M = 1.50 \, \Omega$ is the module's effective electrical resistance at the stated 27°C hot-side condition.
- $I_{\max} = 8.6 \, \text{A}$ is the maximum current; the table identifies it as the current at $\Delta T_{\max}$.
- $Q_{c,\max} = 71.3 \, \text{W}$ is the maximum cold-side heat pumping at $\Delta T = 0$.
- $\Delta T_{\max} = 70.5^\circ\text{C}$ is the maximum face-to-face temperature difference at zero cold-side heat load,

The symmetric model assigns half of the total Joule heating to the object face:

$$\dot{Q}_{J,\max} = \frac{1}{2} I_{\max}^2 R_M = \frac{1}{2} (8.6 \, \text{A})^2 (1.50 \, \Omega) = 55.47 \, \text{W}.$$

At $\Delta T = 0$, the measured cold-side cooling equals the Peltier pumping minus object-face Joule heating, so

$$\dot{Q}_{P,\max} = Q_{c,\max} + \dot{Q}_{J,\max} = 71.3 + 55.47 = 126.77 \, \text{W}.$$

The maximum-current heating-to-cooling slope-ratio prediction is then

$$r_{\text{Laird},\max} = \frac{\dot{Q}_{P,\max} + \dot{Q}_{J,\max}}{\dot{Q}_{P,\max} - \dot{Q}_{J,\max}} = \frac{126.77 + 55.47}{126.77 - 55.47} = \frac{182.24}{71.30} \approx 2.56.$$

5. Part 5.4: Compare the Ratios

The measured ratio $r \approx 2.45$ is around 4% lower than the result obtained from the datasheet values $r_{\text{Laird},\max} \approx 2.56$. It is not expected for these values to agree exactly because the datasheet describes a specific max-current condition, while the measured ratio comes from the apparatus which has different thermal and electrical conditions. $D = 1$ only means that the H-bridge is on continuously. It does not guarantee that $I = I_{\max}$ since the actual current depends on the 12V power supply's current limit, H-bridge wiring and voltage drops, as well as the temp-dependent TEC resistance, while the datasheet max-current point uses $V_{\max} = 13.9 \, \text{V}$ with the hot side at 27°C .

Additionally, the experiment also use PWM rather than the ideal steady DC, operates at finite temperature differences, and has passive heat paths through the supports, leads, and air. Material properties change based on the temperature. These factors make such a small discrepancy reasonable.

6. Part 5.4: Passive Conduction

When the object is hotter than room temperature, passive heat flows from the object to the room and opposes further heating. When the object is colder than room temperature, passive heat flows from the room into the object and opposes further cooling. Approximately symmetric passive conduction therefore reduces the temperature displacement in both directions. However, because it reverses direction with the temperature difference and acts similarly in magnitude for comparable positive and negative differences, it cannot by itself explain unequal heating and cooling slope magnitudes. The primary directional asymmetry in the model comes from Joule heating: it adds to the Peltier contribution during heating but subtracts from the useful Peltier contribution during cooling.